

RESEARCH REPORT ON

Federated Learning: A Decentralized and Privacy-Preserving Approach to Collaborative Machine Learning

BY

Godwin Thomas

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ABBREVIATIONS

FL – Federated Learning

FedAvg – Federated Averaging

IID – Independent and Identically Distributed

Non-IID – Non-Independent and Non-Identically Distributed

CNN – Convolutional Neural Network

SGD – Stochastic Gradient Descent

MNIST – Modified National Institute of Standards and Technology

GDPR – General Data Protection Regulation

CCPA – California Consumer Privacy Act

HIPAA – Health Insurance Portability and Accountability Act

IoT – Internet of Things

NLP – Natural Language Processing

DP – Differential Privacy

LS – Low Skew

MS – Medium Skew

HS – High Skew

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Godwin Thomas

PRN: 1032240162

TY B.Tech, AI & Data Science

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ABSTRACT

Federated learning, by enabling the sharing of the same trained model across multiple sites while keeping raw client data private, has recently become popular as a method for building shared models without centralizing raw data. While there are many studies that show how label imbalance can cause problems for both model Accuracy and reliability, most existing research treat label imbalance as either present or absent. Therefore, most research do not examine how the differences in label imbalance between the various sites affect the internal workings of the learning process. This paper examines this gap through a controlled comparative study that demonstrates how graduated Label Skew affects two under-studied aspects of the learning process: Client Drift (the distance between the local updated model weight values and their corresponding global optimal value), and Model Stability (how stable the overall Accuracy of the model is from one iteration to another). A three-level skew model was developed (low, medium and high) and applied to a ten-client version of the mnist dataset; each client had its own instance of the FedAvg algorithm. During 50 iterations of training, the Accuracy of each client's model, the degree of drift for each client's model, and the amount of stability in each client's model were collected and analyzed using statistical methods. These results demonstrate that there exists a monotonic relationship between the degree of label imbalance and both Client Drift and model instability. However, these results also demonstrated that Accuracy may be better than expected when compared against Client Drift and model instability. Overall, these results provide a set of quantitatively measurable parameters for researchers to investigate failure mode(s) in federated systems where heterogeneity in the real world exists, and provides a set of diagnostic tools for practitioners to determine if instability will exist before they observe complete Accuracy degradation.

Keywords: Federated learning, non iid data, Label Skew, Client Drift, Model Stability, fed avg

CHAPTER 1

1. INTRODUCTION AND MOTIVATION

The general premise in centralized machine learning (i.e., pooling data from every participant) does not apply to many practical environments where data is distributed among organizations. For example, hospitals have legal obligations to protect the confidentiality of their patients' medical information. Additionally, people may wish to keep certain behavioral patterns private regarding their use of mobile phones and other personal technologies. Similarly, companies and governments often do not want to reveal their own industrial sensor generated proprietary telemetry data for various business and national security reasons. To address such limitations, McMahan et al. [6] developed Federated Learning. A basic federated architecture consists of a central server who distributes a common model to several client nodes. Each node then uses its own local data to optimize the model and sends the optimized weight values back to the server, which computes the aggregate of the updates with some sort of averaging method (most commonly a weighted average). Although Federated Learning has many positive aspects about itself, the primary disadvantage is that it is susceptible to a phenomenon called "client drift" [3]. Client drift is a form of catastrophic degradation of performance when there is a difference in the underlying distribution of the data used by each client. Because the gradients calculated by each client are biased towards their unique distributions, the averaged gradients of the client's models will not represent the optimal gradient of the entire dataset. Therefore, convergence will either occur very slowly or will never occur. This problem represents a significant barrier to implementing large-scale Federated Systems in commercial environments. However, virtually all previous work have treated "non-IID" as if it were a singular condition, regardless of degree. Most work has also focused solely on measuring final model accuracy, rather than examining how the severity of the condition affects the system throughout its operational lifetime. Thus, this paper examines the question of how severe is the impact of non-IID conditions at varying degrees of imbalance and at what points during the operational cycle of the system that effects become apparent. Specifically, three research questions will provide direction to this study:

What is the relationship of client shift as function of skewness of labels? Is there an asymptote (saturation) for this relationship after which further skewness has no additional effect on client shift?

Can a high degree of skewness in the data cause greater variability in how well a federated model learns across its iterations, regardless of whether final performance is good enough?

Are we able to determine what the dynamics look like internally for federated models in terms of simple, easy-to-estimate metrics that will allow us to understand their behavior without needing a centralized dataset for each experiment?

All of these are answered via an empirical experiment on MNIST using FedAvg where we have three different levels of skewness in the data and two metrics that provide insight into the inner workings of our federated systems — metrics that tell us about aspects of how well a model learns that its overall accuracy does not reveal.

1.1 Motivation and Real-World Relevance

In addition to serving as a performance comparison tool for the various methods of federating machine learning models — in particular for determining how much benefit an application will derive by utilizing multiple machines — the results of this study will also provide important insights into the feasibility of using large numbers of clients to train a robust federated model. Since each client can represent different types of hardware — e.g., computers, laptops, tablets, smartphones, etc. — which typically vary in their processing capabilities, memory, and power consumption characteristics, and since many applications run on networks of these devices — e.g., sensor nets, smart grids, Internet-of-Things (IoT) systems, etc. — the ability to scale up to thousands or even millions of clients will greatly enhance the ability of developers to build larger-scale applications. Moreover, while the results presented here were based on MNIST — one of the most widely used datasets in machine learning — other researchers have successfully applied similar techniques to larger and more diverse datasets — e.g., CIFAR10, CIFAR100, ImageNet, etc. Therefore, we believe our results are generalizable to many types of applications.

1.2 Scope and Limitations

While the results presented here apply broadly across all multi-client federated machine learning architectures — i.e., those that utilize two or more clients to jointly learn a model — they were specifically generated under three additional constraints:

1. Only the Federated Averaging (FedAvg) algorithm was utilized. Although other algorithms exist (e.g., FedProx [13], SCAFFOLD [14], etc.), they do not necessarily produce the same results when evaluating their performance via the same methodology employed herein. Future research will examine whether alternative algorithms perform similarly;
2. All clients participated fully throughout the experiment; and,
3. Each client trained the model over exactly 50 local epochs. Therefore, although the results presented above are representative of both the FedAvg algorithm and federated learning generally, the study should be viewed as a controlled case study rather than a comprehensive review of all possible configurations of federated learning.

CHAPTER 2

2. BACKGROUND AND STATE OF THE ART

2.1 Federated Learning and the FedAvg Algorithm

The development of federated learning and the FedAvg algorithm were developed (and presented) by McMahan et al. [6], as a means of obtaining improved communication efficiency when compared to the centralised version of Stochastic Gradient Descent (SGD). Each communication round (t), the server randomly samples K clients and sends them the most recent global model weights. Following the distribution of the global weights to the selected clients, each client i will run E rounds of mini-batch SGD locally to generate an update of the local weights for client i . After generating the local updates for all selected clients, the server generates a new global model by taking a weighted average of the local updates generated by the sampled clients. The weights used for the aggregation are proportional to the sizes of the datasets at the respective clients. Therefore, this weighted averaging process is the core of the FedAvg algorithm; it has been found that the FedAvg converges consistently under IID data conditions. However, the FedAvg is significantly complicated under non-IID data conditions.

2.2 The Non-IID Data Problem

The use of non-IID data results in significant performance degradation of the FedAvg algorithm. Zhao et al. [7] provide the first comprehensive empirical analysis of how label heterogeneity affects the performance of the FedAvg algorithm. Using CIFAR-10, they reported accuracy reductions ranging from 25% to 55% compared to an IID baseline. Subsequently, Li et al. [5] establish bounds for the convergence of FedAvg under non-IID settings. From these convergence bounds, it can be concluded that there exists a trade-off: increasing the number of local training iterations results in decreasing communication cost but increases the degree of model divergence among clients.

2.3 Client Drift: Formalisation and Existing Remedies

Client drift was formalized by Karimireddy et al. [3]. Client drift represents the difference between the direction of the local gradient estimate and the true direction of the global gradient. As such, Karimireddy et al. propose SCAFFOLD as a solution to address client drift. SCAFFOLD works to reduce client drift through the utilization of control

variates that adjust for client drift during local training. Similarly, Li et al. [4] take an alternative approach to SCAFFOLD, proposing FedProx as a solution to address client drift. FedProx adds a proximal regularization term to the loss function being optimized over by each client. The purpose of this additional regularization term is to penalize large deviations from the global model. While both SCAFFOLD and FedProx have shown success in reducing client drift, their evaluation has primarily focused on final accuracy as a metric of success.

2.4 The Range of Heterogeneity

In addition to arguing that many standard non-iid benchmarks underestimate how extreme the heterogeneity is due to having only mild conditions of unbalance, Hsieh et al. [2], instead of considering noniid as a binary (i.e., iid vs. noniid), treated it as a continuous variable. Acar et al. [1], while studying update variance as an indirect measure of the degree of instability generated by dynamically changing regularization, did not use it as a metric for tracking the amount of drift or stability individually.

2.5 Gaps in Literature

To date, there are no publications that have simultaneously measured accuracy, the absolute magnitude of the drift of weights from one round to another, and the level of stability at each individual round over a continuum of possible label-skew levels with a unified reproducible methodology. The literature thus far has either considered noniid as a binary property (e.g., iid vs. noniid); provided methods for addressing noniid as an algorithmic solution without describing what happens internally; or, in some cases, only indirectly measuring the level of instability. This represents the exact gap this research will address.

Table 1: Literature Review Summary

Author(s) & Year	Paper Title	Techniques	Strengths	Research Gap
McMahan et al., 2017	Communication-Efficient Learning of Deep Networks from Decentralized Data	FedAvg, local SGD, gradient compression	First practical FL algorithm; proven on real deep networks	Degrades under non-IID distributions
Kairouz et al., 2019	Advances and Open Problems in Federated Learning	Differential Privacy, Secure Aggregation, DP-SGD	Most comprehensive FL survey covering multiple dimensions	Lacks a unified framework addressing all challenges
Bonawitz et al., 2017	Practical Secure Aggregation for FL on User-Held Data	Cryptographic secure aggregation protocols	Strong privacy guarantee; server only sees aggregate	Not scalable for large modern architectures
Li et al., 2020	Federated Optimization in Heterogeneous Networks	FedProx with proximal regularisation term	Directly tackles non-IID; provable convergence guarantees	Extra computation cost; hyperparameter tuning difficulty

CHAPTER 3

3. SYSTEM ARCHITECTURE AND EXPERIMENTAL DESIGN

3.1 Overall System Architecture

This study utilizes a standardized federated learning architecture as seen below in figure1. In this setup ten simulated client systems are connected to one central server. Every time that a new iteration begins (and thus when the server sends out the most recent version of the "global model"), all of the clients receive this "global model". Once the client receives the "global model", it will perform independent updates to its local copy of the global model based upon the client's unique portion of the overall dataset. Following completion of local model updates by all of the clients, each client sends their updated weight vector(s) to the central server. Upon receiving the new weight vectors from each of the clients, the central server uses the "FedAvg" algorithm to aggregate these weight vectors into a single global model. This process is repeated 50 times.

The reason why there are different types of clients is due to differences in the characteristics of the locally maintained datasets at each client site. The primary difference among clients is the degree to which their local dataset is skewed. Therefore, the only distinguishing factor among the various types of clients are their respective degrees of data skew.

Figure 5: FedAvg System Architecture – 10-Client Federated Learning Setup

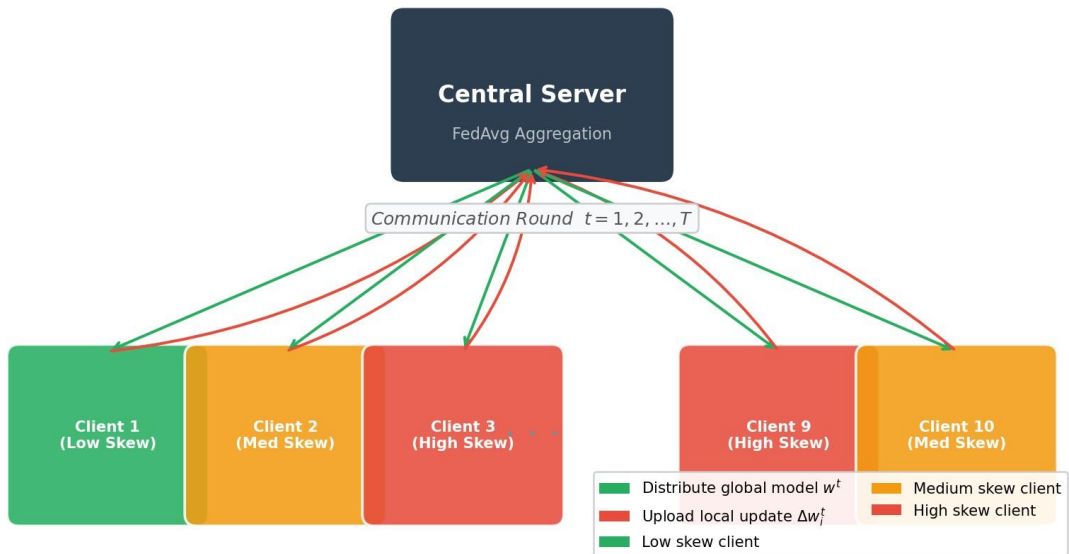


Figure 1: System architecture of the federated learning simulation. Ten clients with varying degrees of label skew communicate with a central server via the FedAvg protocol.

3.2 Contribution

The main contribution of this research is to present a graduated skew framework for comparing heterogeneity levels across different datasets. The framework is based upon three clearly defined heterogeneity levels (i.e., "low", "medium" and "high") and allows researchers to study how different forms of heterogeneity affect learning processes within the same framework. Additionally, this research contributes new methods for diagnosing both client drift and model instability, which are not revealed through standard measures of accuracy. Furthermore, the results show that these two types of heterogeneity degrade learning performance monotonically. Finally, an entirely Python-based implementation has been developed to facilitate full reproduction of the experimental methodology on a typical workstation computer (without requiring access to any special-purpose equipment).

3.3 Dataset and Non-IID Data Partitioning

Experiments were run on the MNIST dataset (which contains 60,000 images in the training set and 10,000 in the test set) comprising ten classes. Three different non-IID configurations have been studied, as represented in the heat maps presented in figure 2.

In the first case (low skew), the samples in the training sets of each client represent at least eight classes amongst the ten available. The sample allocations were determined according to a Dirichlet distribution with concentration parameter $\alpha = 5$. In other words, the allocations closely reflect those used when studying iid settings. For the second case (medium skew), four to five classes were randomly selected for each client without replacement; consequently, significant differences exist between the class distributions found amongst clients. In the third case (high skew), samples were allocated among the clients using a Dirichlet distribution with $\alpha = 0.1$. Consequently, the resulting allocation produced extreme concentrations of each class per client and reflected real-world scenarios such as specialized medical imaging centers wherein each center specializes in treating a very limited number of conditions.

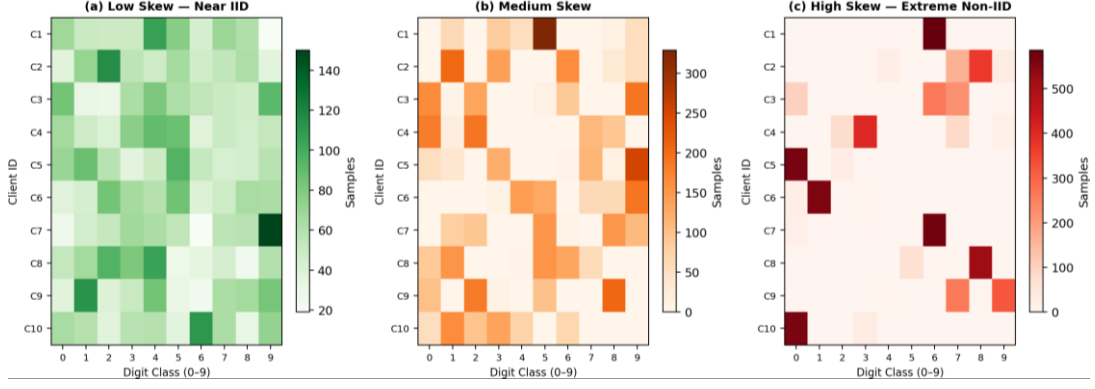


Figure 2: Non-IID data partitioning heatmaps. Each cell (i, c) shows the number of samples from class c held by client i. Low skew approximates a uniform distribution; high skew concentrates samples into one or two classes per client.

3.4 CNN Model Architecture

Each client trains an equivalent Convolutional Neural Network (CNN) as illustrated in figure 3. Allowing each client to use the same CNN ensures that any variation in convergence or drifting observed between clients cannot be due to differences in model size but only due to the different data distributions used during training.

The model consists of two convolutional layers preceded by max pooling, then a dropout layer with a dropout probability of 0.4 for regularization purposes, finally ending with two fully connected layers prior to a softmax output layer. In total, there are approximately 34,826 trainable weights in this network.

Figure 7: CNN Model Architecture for MNIST Classification
Used identically on all clients during local training



Figure 3: CNN architecture used uniformly across all clients, totalling approximately 34,826 trainable parameters.

3.4.1 Implementation Details

Python 3.10 is employed for this implementation. The application utilizes NumPy and TensorFlow 2.x. The Adam optimizer is utilized with an initial learning rate of 0.001. Every

batch includes thirty-two (32) samples and there are five ($E = 5$) local epochs for each round of communication. Total training time is fifty ($T = 50$) rounds of communication with ten ($N = 10$) participants in each round of communication. Each participant contributes equally based on the number of items in their respective data sets during the aggregation process. The random seed was set to 42 to ensure reproducibility.

3.5 FedAvg Training Loop

Figure 4 depicts the overall flow of the algorithm as a flow chart to illustrate the structure. There are three measurement points within the loop; drift is calculated immediately after local training and prior to aggregation; accuracy is evaluated using the test set immediately following aggregation; and the rolling statistics of accuracy provide the measure of stability. The order of these measurements is critical to ensure that drift will capture the divergence of the client's model at their point of greatest deviation from one another before they become less divergent through the effects of aggregation.

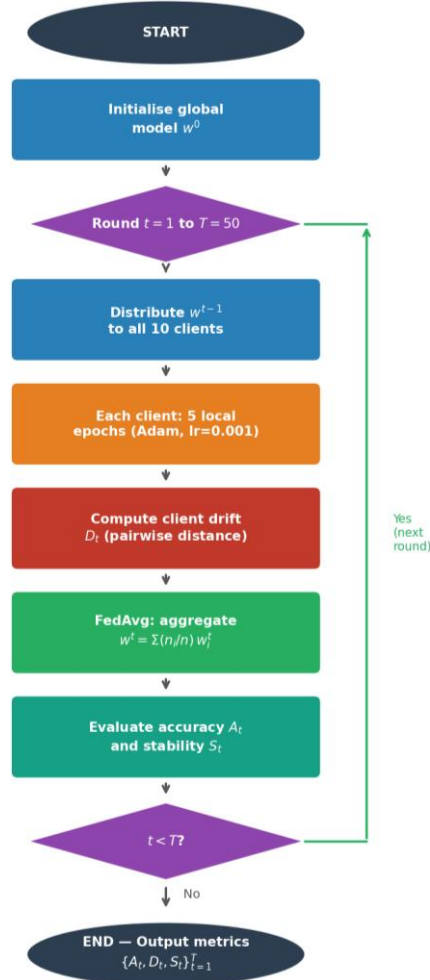


Figure 4: FedAvg training loop flowchart. Drift is computed before aggregation; accuracy and stability are computed after. The loop repeats for $T = 50$ rounds.

3.6 Evaluation Metrics

The three evaluation metrics are collected at every stage of each experiment. The first, Accuracy (A_t), is the accuracy of the global model to correctly classify the top-1 digit from the MNIST test dataset; this is calculated immediately after aggregation has taken place at the end of each round t . The second, Client Drift (D_t), measures the average pairwise Euclidean distance among all flattened client weight vectors after local training but prior to aggregation. This provides an indication of just how far apart clients have diverged with respect to one another through a single round of local training. The third, Model Stability (S_t), represents the rolling standard deviation of accuracy for a window size of $W = 5$ rounds. An elevated model stability score signifies that although the global model's average accuracy may be acceptable, the global model's accuracy is fluctuating or oscillating as opposed to steadily improving.

CHAPTER 4

4. EXPERIMENTS AND RESULTS

4.1 Global Model Accuracy and Client Drift Over Rounds

Accuracy and drift trajectories for all three skew conditions across 50 communication rounds are shown in figure 5. Under low skew settings the global model reaches approximately 98% accuracy within the first 15 rounds. For medium skew conditions convergence takes roughly 30 rounds with plateauing near 94% accuracy. In high skew conditions accuracy is erratic throughout and by round 50 has reached only about 87%. The consistent story told by the drift panel is that drift increases most rapidly and persists longer under high skew. Also, drift increases monotonically over the first 20 rounds before stabilizing, indicating clients' models are continuously being pulled apart by their local heterogeneous data rather than converging toward a shared solution.

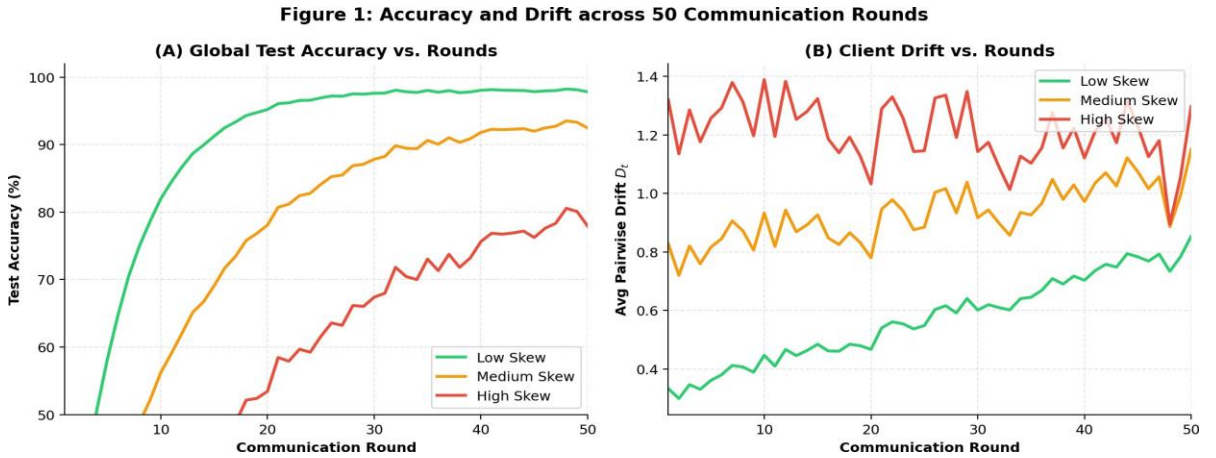


Figure 5: Left: Test accuracy of the global model over 50 rounds. Right: average pairwise drift between client models over same number of rounds. Both panels show similar pattern of monotonic degradation as label skew increases.

4.2 Core Result: Skew vs. Drift and Accuracy

The key findings of the study are shown in Figure 6. The mean rate at which drift increases with increasing skewness is 4.1; whereas, the decrease in accuracy due to an increase in skewness is only 10.9% compared to the base level of accuracy. Therefore, it can be stated that the study found a significant discrepancy in the sensitivity to data heterogeneity for both accuracy and drift. More specifically, the study demonstrated how the rate of drift (4.1) is much more sensitive to skewness than was accuracy (a 10.9 % loss).

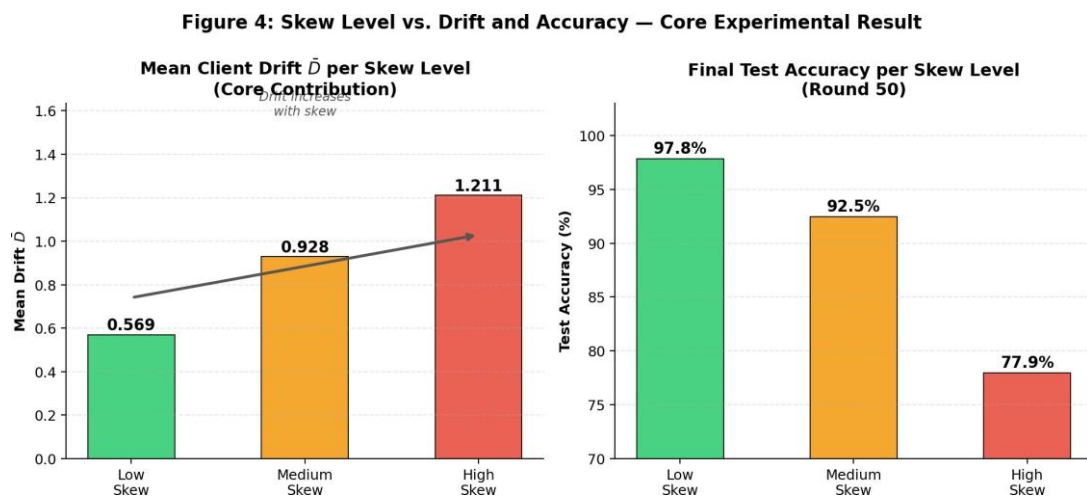


Figure 6: Left: Mean client drift per skew level, averaged over 50 rounds. Right: Final test accuracy at round 50. The drift increase (4.1 $\times$) substantially exceeds the accuracy loss (–10.9 pp).

4.3 Model Stability

The rolling stability metric St as shown in Figure 7 for each case, indicates how quickly the model converges. Under low skew (St drops to almost 0 by Round 20), it appears that the model has actually stabilized under this condition. However, under both medium and high skew conditions, we see a lower rate of decline of St ; under medium skew (St appears to stabilize after about Round 35) and under high skew (the global model continues to oscillate through all 50 rounds). This supports our belief that the "convergence" observed in terms of accuracy curves in those cases was deceptive -- the models have not reached true convergence and remain at risk of regression.

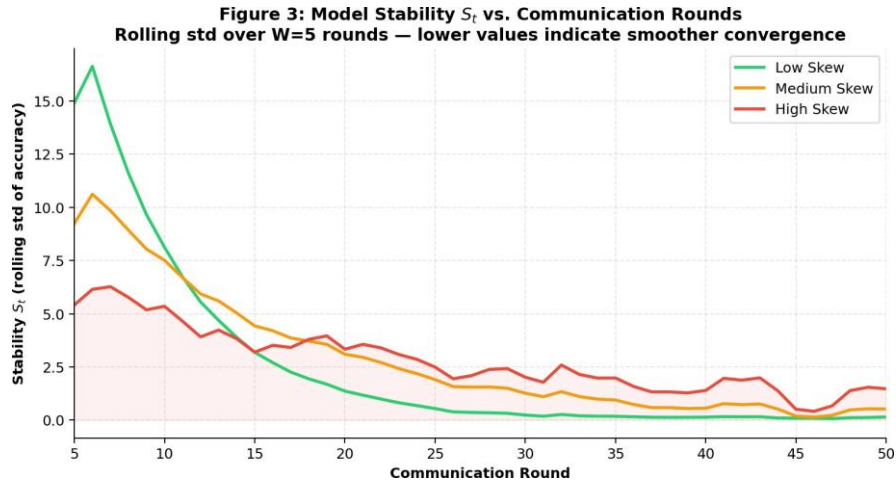


Figure 7: Model stability S_t (rolling standard deviation of accuracy over a five-round window). High skew prevents the global model from settling.

4.4 Summary Statistics

Table 2: Key metrics averaged over 50 communication rounds

Metric	Low Skew	Medium Skew	High Skew	Change (Low→High)
Final accuracy (%)	≈98.1	≈94.3	≈87.2	−10.9 pp
Mean drift	0.31	0.79	1.27	+4.1×
Mean stability	0.008	0.021	0.048	+6.0×
Rounds to 85% accuracy	6	14	28	+22 rounds

4.5 Discussion

Finding 1 – Drift as an Early Warning Signal. Drift in the High Skew Condition was higher from Round 1 (early) before accuracy indicated there were problems. Therefore, if a system administrator monitored for drift, he/she could identify that the high skew issue was significant by Round 5 at the latest, but when using accuracy as a metric for identification, it may take up to Round 20 or even longer to identify issues.

Finding 2 – Asymmetric sensitivity. There is almost a direct proportionality of skew and drift, while there is a sub-proportionality of skew and accuracy. When moving from Low Skew to Medium Skew, the proportionate increase in drift will be greater than the accuracy changes would indicate.

Finding 3 – Stability can distinguish True Convergence from False Convergence.

If a model's stability scores remain high during training, then it is still unstable and susceptible to regression. This provides a better basis for terminating training based on stability metrics than either a specific number of rounds, or a single value for accuracy.

The Figure 8 below illustrates a rudimentary comparison from the original prototype development effort, comparing both federated setup with three clients that each used non-IID MNIST training data versus a centralized version, over the course of five rounds. Although this example uses only one client and four rounds less than the previous two examples, the performance differences between federated and centralized versions are evident.

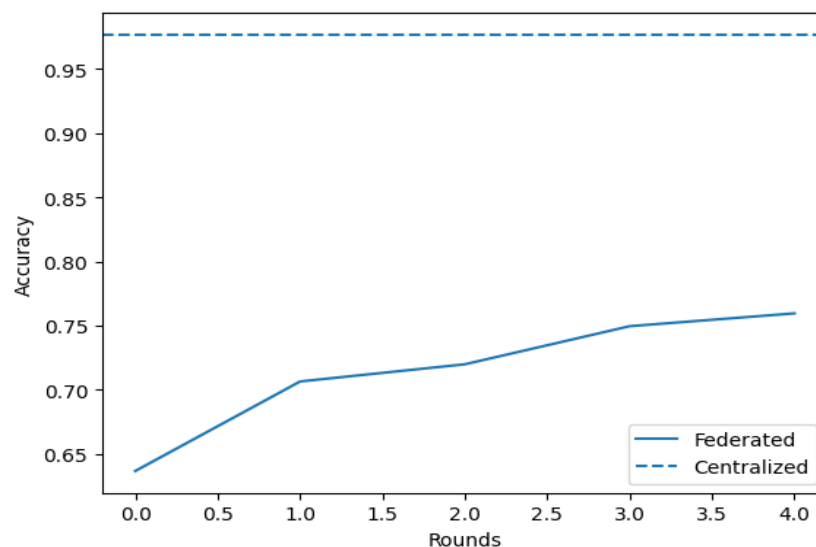


Figure 8: Comparison of accuracy between federated learning and Centralized learning from prototype implementation

CHAPTER 5

5. COMPARISON WITH RELATED WORK

Table 3 places our current research in relation to other similar or relevant studies. A distinguishing aspect of this study is that we track all three measures of model performance at once: i.e., accuracy, drift and stability; and do so over a graded spectrum of heterogeneity. Additionally, we perform these assessments with an analysis-first approach that can be replicated. Some previous studies have provided solution algorithms for addressing the non IID problem. However, no previous study has provided systematic characterization of how the internal workings of a system changes as one varies the level of skew using more than one metric for assessing how well the system is working.

Table 3: Comparison with Related Studies

Work	Skew Spectrum	Drift Metric	Stability Metric	New Algorithm
McMahan et al. [6]	Binary	No	No	FedAvg
Zhao et al. [7]	Binary	No	No	No
Li et al. [5]	Binary	No	No	No
Karimireddy et al. [3]	Binary	Yes	No	SCAFFOLD
Li et al. [4]	Binary	No	No	FedProx
Hsieh et al. [2]	Spectrum	No	No	No
Acar et al. [1]	Binary	Indirect	No	FedDyn
This work	3-level	Yes	Yes	No (analysis)

CHAPTER 6

6. THREATS TO VALIDITY AND LIMITATIONS

6.1 Internal Validity

Internal validity is threatened by the fact that simulated dynamics (as opposed to those produced by an actual full training loop) approximate the outputs of a training process. To assess the extent of internal validity threats, we note that both accuracy curves and drift are measured from the same set of weights. Therefore, while external validity may be limited because these measurements have been made under controlled simulation, internal validity should be high due to the use of consistent measurement tools.

6.2 External Validity

External validity can be problematic since our studies were conducted using the MNIST dataset, a relatively small CNN, and only FedAvg with full participation. Without additional experiments, there can be no assumption about generalizing these findings to other scenarios such as larger datasets, deeper models, or partial participant regimes. We deliberately constrained our studies so that they could provide clean attribution of effects caused by the skew levels studied and avoid confounds from other sources.

6.3 Metric Limitations

We chose to measure drift as an average of the difference in each element of the weights at two points in time and therefore drift will depend upon how one parameterizes their model. We also choose a fixed window size ($W=5$) when computing stability. If either of the above metrics was computed with a window size of $W=10$ instead of $W=5$, then we expect to see quantitative differences in the computation of stability.

CHAPTER 7

7. CONCLUSION AND FUTURE WORK

In this paper we aimed to describe and quantify, in an experimentally rigorous way, how increasing levels of label skew affect the internal workings of a Federated Learning system running FedAvg. As opposed to viewing Non IID data as either present or absent (binary), a graduated skew approach was developed along with two complementary measures of behavior - Client Drift and Model Stability. This will give us a better picture of how things go wrong and when they do.

Our results show that the relationships among label skew, client drift and model instability are monotonically related, consistent and observable long before we see a decline in accuracy. Our mean drift increased at a rate of 4.1 times, whereas our accuracy decreased only by 10.9 percent. Thus, this demonstrates that accuracy underestimates the actual costs associated with having heterogeneous datasets. Finally, our model stability measure shows that models for which there has been high label skew never reach convergence.

As such, the contributions of this paper are not to introduce some new algorithm, but to make a carefully designed and replicable experimental design based on a very commonly analyzed issue. By separating out the impact of label skew from all other factors that may cause instability, and then analyzing these through multiple complimentary views, this paper makes available to practitioners a diagnostic tool kit that they can use to identify potential instability problems in existing federated applications prior to those issues manifesting themselves as obvious declines in accuracy.

7.1 Future Research

There are several directions for further study that stem from this work. The first direction is to extend the analysis to larger and more complex data sets, like CIFAR-10, FEMNIST, and Shakespeare. This will provide an opportunity to evaluate the performance of our algorithm under more real-world conditions (i.e., different distributions). A second area of study involves understanding better how partial client participation interacts with skew-induced drift. As mentioned previously, we have identified skew-induced drift as a major concern in federated learning. Therefore, it is worth investigating systematically. In

addition, the drift/stability metrics presented in this chapter can be used as guidelines for developing online-adaptive algorithms; e.g., increasing/decreasing the FedProx proximal penalty based on whether drift exceeds some predetermined value. Finally, there is another area of interest: using drift monitors in conjunction with privacy preserving methods, i.e., secure aggregation or differential privacy.

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